Grading Handwritten Answers with Vision-Language Models



Alfiia Ziganshina

AI Seminar — Topic proposal (draft, idea stage)

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JOHANNES KEPLER UNIVERSITY LINZ

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Agenda

1. **Why**
   Motivation, how STACK works, what exists
2. **What**
   Three candidate tasks and one idea per task
3. **Compare**
   Tasks side by side: data, difficulty, value
4. **Resources**
   Datasets, models on Hugging Face, key papers
5. **How**
   Approach, idea backlog, risks
6. **Next**
   Draft research question, plan, questions

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Motivation

- Teachers grade handwritten worksheets and exams by hand — **slow and repetitive**
- ETH Zurich's **STACK** grades math by meaning (2(x+1) = 2x+2), but only for **typed** answers
- Vision-language models (VLMs) can now read handwriting and judge answers
- Open question: how far can **open, fine-tuned** models get, and where do they fail?

95%

rubric-item accuracy of the best VLM grader on university STEM exams

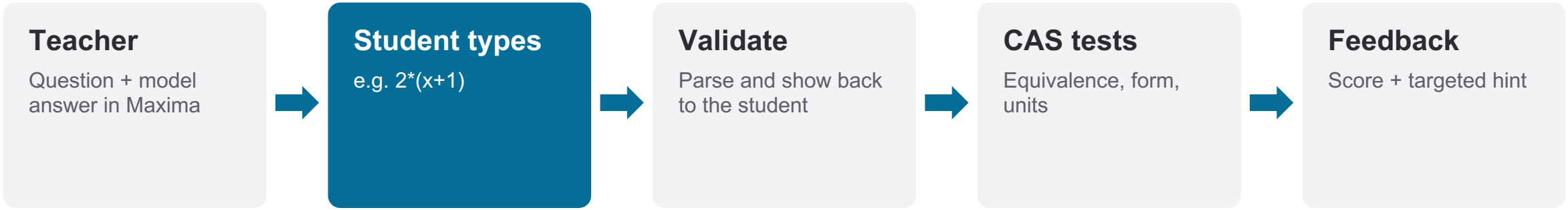
87%

of one model's grading errors came from misreading the handwriting

Levine et al., AIED 2026 (arXiv 2605.19043)

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How STACK works, and what we would change



OUR CHANGE

Student types → Student writes on paper, a model reads it

Everything after reading can stay STACK-like: a deterministic check by meaning, plus feedback.

Open idea: show the student the model's reading before grading, as STACK's validation step does (cf. ITiCSE 2026 "correcting transcripts" study).

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What already exists

TYPED + COMPUTER ALGEBRA

STACK (ETH), Numbas, WeBWorK, PrairieLearn

Checks answers by mathematical meaning; randomised questions; instant feedback.

Limit: students must type their math.

HANDWRITTEN + AI ASSIST

Gradescope (Turnitin)

AI groups similar handwritten answers; the instructor grades each group once.

Limit: a human still decides; fixed-template PDFs only.

RESEARCH: VLM GRADERS

Gemini, GPT, Qwen-VL (2025–26)

Read the scan and judge it directly against a rubric or answer key.

Limit: no symbolic check; errors mostly from reading.

Gap: handwritten input combined with a STACK-like meaning check, measured end to end.

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Topic: three grading tasks, from easy to hard

1 EASY

Fill-the-gaps worksheets

Printed text with gaps; students write in the missing words.

Model task: Read each gap, compare with the answer key (spelling variants).

Hard part: Small handwriting inside printed text

2 MEDIUM

Free-form handwritten text

Essays in any language (most data is English).

Model task: Transcribe the full page, then score it with a rubric.

Hard part: Messy handwriting, crossed-out and corrected words

3 HARD

General math

Formulas and derivations.

Model task: Formula → LaTeX, then check each step by meaning, like STACK.

Hard part: 2-D notation; a typo is not always a real error

Scope: pick about two tasks, based on which public datasets exist. Building our own dataset is out of scope.

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Idea 1: fill-the-gaps worksheets

1 EASY

EXAMPLE

Printed: “Paris is the ____ of France.”

Written: ***capitol***

Key: capital

Output: wrong (spelling) — or accept? A policy decision for the teacher.

Pipeline	Find gaps from the template → crop → read the word → compare with the key
Data	Grabowski answer boxes, IAM words / lines (proxy, English)
Models	TrOCR (small), PaddleOCR-VL, Qwen-VL
Metric	Per-gap accuracy; rate of wrongly accepted answers
Risk	No real worksheet dataset: evaluation only on proxies

Neural-network part: fine-tune a small reader on word / answer-box crops.

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Idea 2: free-form handwritten text

2 MEDIUM

EXAMPLE

Input: one handwritten essay page, incl. crossed-out words

Step 1: transcript of the page

Step 2: rubric scores, e.g. content 4/5, language 3/5, structure 4/5

Output: scores + short feedback

Pipeline	Page → transcription → LLM scores with rubric; vs. one VLM end to end
Data	GCSE double-marked (32.5k), VisEssay (13k+), BRESSAY
Models	chandra-ocr-2, olmOCR 2, Qwen-VL + LLM scorer
Metric	Character error rate; quadratic weighted κ vs. human marks
Risk	Dataset access and licences; subjective scores

Neural-network part: fine-tune the reader on essay pages, or a VLM scorer end to end.

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Idea 3: general math

3 **HARD**

EXAMPLE

Written: $(x^2 \cdot \sin x)' = 2x \cdot \sin x + x^2 \cdot \cos x$

Read: LaTeX for each line

Check: SymPy: each line equivalent to the previous one?

Output: correct, or “first error in line 3”

Pipeline	Line detection → LaTeX → SymPy check per step → feedback
Data	MathWriting, HME100K, CROHME (reading); FERMAT (grading)
Models	Qwen-VL fine-tuned; TexTeller as baseline
Metric	Exact formula match; error detection F1; error location
Risk	LaTeX that SymPy cannot parse; multi-line layout

Neural-network part: LoRA fine-tune Qwen-VL on handwritten formulas → LaTeX.

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Comparing the three tasks (first impression)

Criterion	1 Fill the gaps	2 Free text	3 Math
Public data	Low — proxies only	Medium — English sets	High — reading + grading
Difficulty	Low	Medium	High
Neural-network work	Medium — small reader	High — page reading	High — formula reading
Close to STACK	Medium — answer key	Low — rubric	High — meaning check
Value for teachers	High — common format	High — slow to grade	High — slow to grade
Main risk	No real worksheets	Access; subjective scores	Parsing multi-line math

First impression: Math as main task; second task to decide together.

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Public datasets per task

Task	Dataset	Contents	Source
1 Fill the gaps	Grabowski exam benchmark	3,141 handwritten answer boxes, 61 exams	arXiv 2606.11477
1 Fill the gaps	`Teklia/IAM-line`	English handwritten text lines	Hugging Face
1 Fill the gaps	`fhswf/german_handwriting`	German handwriting samples	Hugging Face
2 Free text	GCSE double-marked benchmark	32.5k responses incl. handwritten essays	arXiv 2606.24973
2 Free text	VisEssay	13k+ graded handwritten essays (no public download)	ACM 2022
2 Free text	BRESSAY	1,000 handwritten essay pages (Portuguese)	2025 study
3 Math	`ai4bharat/FERMAT`	2.2k solutions with labelled errors	Hugging Face
3 Math	`andito/mathwriting-google`	253k handwritten formulas → LaTeX	Hugging Face
3 Math	`ConstantHao/HME100K`	99k photographed formulas	Hugging Face
3 Math	CHECK-MAT	122 exam solutions with expert grades	arXiv 2507.22958

Proposal: 3 Math (most public data, closest to STACK) + one of 1 Fill the gaps (easy baseline) or 2 Free text (large English essay sets). Datasets need not be German.

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Candidate models on Hugging Face

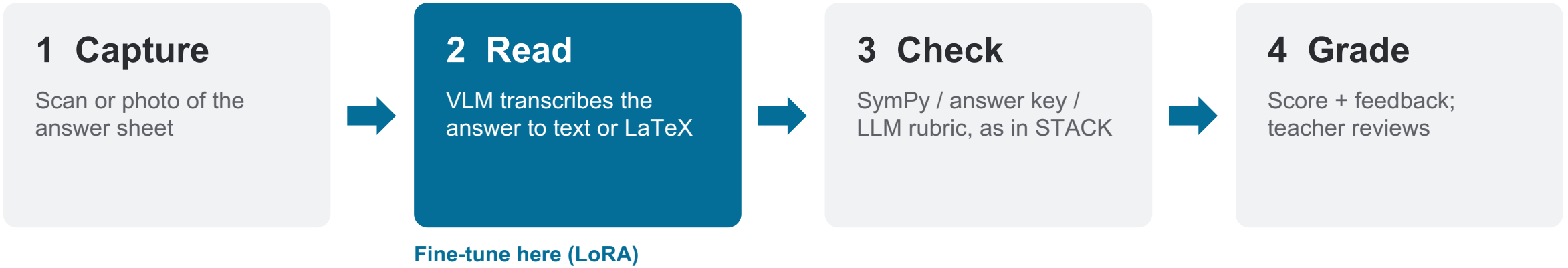
Hugging Face id	Type	Tasks	Role in the project
`Qwen/Qwen2.5-VL-7B-Instruct`	General VLM	1 · 2 · 3	Main fine-tuning base (5.4M downloads)
`Qwen/Qwen3-VL-4B-Instruct`	General VLM	1 · 2 · 3	Smaller, newer base; fits one GPU
`deepseek-ai/DeepSeek-OCR-2`	OCR VLM	1 · 2	Zero-shot reading baseline
`datalab-to/chandra-ocr-2`	OCR VLM	1 · 2	Reported strongest on handwriting
`allenai/olmOCR-2-7B-1025`	OCR VLM	2	Open OCR, Qwen-based
`PaddlePaddle/PaddleOCR-VL-1.6`	OCR VLM	1 · 2	Small (~1B), top on OmniDocBench
`OleehyO/TexTeller`	Formula OCR	3	Formula image → LaTeX baseline
`microsoft/trocr-base-handwritten`	Handwriting OCR	1	Small classic baseline (English)
`fhswf/TrOCR_german_handwritten`	Handwriting OCR	1 · 2	Small German baseline (optional)

Key papers

Task	Paper	Venue	Why it matters
3	Nath et al. — Can Vision-Language Models Evaluate Handwritten Math? (FERMAT)	ACL 2025	Benchmark: detect, locate, fix errors
3	Levine et al. — Automated Grading of Handwritten Mathematics Using Vision-Capable LLMs	AIED 2026	95% per rubric item; reading is the bottleneck
3	VEHME — A VLM for Evaluating Handwritten Mathematics Expressions	EMNLP 2025	Fine-tuned grader (SFT + RL)
3	MathWriting — Dataset for Handwritten Math Expression Recognition	Google 2024	Training data for formula reading
3	Evaluating AI Grading on Real-World Handwritten College Mathematics	arXiv 2026	3,945 calculus answers vs. TA scores
1	Grabowski — Towards Fully Automated Exam Grading	arXiv 2026	98.4% on handwritten answer boxes
2	LLM Performance on a Real, Double-Marked GCSE Benchmark	arXiv 2026	Handwritten essays, two human markers
2	Evaluating Austrian A-Level German Essays with LLMs	arXiv 2026	German essays; handwriting OCR failed
2	EssayJudge — Multimodal Benchmark for Essay Scoring	ACL Find. 2025	Multimodal LLMs scoring essays

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Proposed approach: read, then check like STACK



EXPERIMENTS

- Zero-shot baselines: OCR models vs. general VLMs
- Fine-tune Qwen-VL on MathWriting / HME100K
- Read + symbolic check vs. VLM grading directly
- Optional: ensemble of two readers

METRICS

- **Reading:** character error rate, exact formula match
- **Grading:** accuracy and F1 vs. human labels
- **Agreement:** Cohen’s κ with teachers
- **Errors:** share caused by reading vs. judging

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Idea backlog (brainstorm)

1

Confidence routing

Unsure answers go to the teacher; sure ones are graded automatically.

2

Two-reader ensemble

Two models read the same answer; if they disagree, flag it.

3

Crossed-out text

Detect strike-throughs before reading (handwritten_cross-outs data).

4

Step-level checking

Check each derivation line, so one typo does not fail everything.

5

Feedback, not only scores

Explain the error to the student in one sentence.

6

Fairness

Compare accuracy across handwriting styles and writers.

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Risks and open questions

RISKS

Risk	Mitigation
Dataset access / licences	Check early; keep a backup dataset per task
GPU compute	Smaller models (Qwen3-VL-4B), LoRA, 4-bit
No own data allowed	Proxy datasets; be explicit about limits
Scope too large	Two tasks max; math first
Closed APIs change	Use them only as reference points

OPEN QUESTIONS

- Which two tasks?
- Is there GPU access at JKU?
- What should the seminar paper be: survey, experiments, or both?
- Is non-German data fine for all tasks? (yes, for now)
- Should the seminar topic stay open for a later thesis?

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Research question and plan

RESEARCH QUESTION (DRAFT)

Does fine-tuning an open VLM on handwriting improve grading accuracy when its reading is checked STACK-style?

Hypothesis: fine-tuned reading + symbolic check beats a zero-shot VLM that grades directly.

PROPOSED PLAN

1. **Nov** Literature: 5 papers in depth, ~20 abstracts; send paper list
2. **Nov–Dec** Confirm the two tasks; zero-shot baselines on FERMAT / MathWriting
3. **Dec–Jan** Fine-tune, symbolic check, evaluation
4. **Jan** Seminar paper and talk

QUESTIONS FOR TODAY

- Math + Fill the gaps, or Math + Free text?
- Is there GPU access at JKU for fine-tuning?

JOHANNES KEPLER
UNIVERSITY LINZ

Thank you!



Questions and feedback?
Alfiia Ziganshina — AI Seminar

JOHANNES KEPLER
UNIVERSITY LINZ
Altenberger Straße 69
4040 Linz, Austria
jku.at